SCI-INTEL: Acquisition-Aware Retrieval and Quality-Risk Curation of Scientific Microscopy Images

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***Abstract*—Micrographs in scientific repositories look different depending on instrument, detector, and accelerating voltage, which weakens similarity search and makes quality control slow. We present SCI-INTEL, a locally validated platform for retrieval and quality-risk curation of scanning electron microscopy (SEM) images. It keeps two separate representations, a frozen DINOv2 ViT-S/14 for image-derived quality screening and an acquisition-aware supervised-contrastive adapter for retrieval, and joins them through deterministic evidence retrieval, rule-based suggested actions, uncertainty-aware abstention, and an auditable human-review workflow rather than a learned fusion. On an instrument-held-out split of the HCCI cast-iron collection (212 queries, Zeiss Gemini), the adapter reduced the observed acquisition-geometry similarity gap from 0.2016 to 0.0681 (66.23%; paired d_z=2.19, p=5.03×10^{-36}, N=210 queries) and raised Recall@5 from 0.9858 to 0.9921 under an unmasked cross-acquisition protocol, while lowering 11-class artifact macro-F1 from 0.6837 to 0.6323. On a controlled synthetic-degradation benchmark, frozen DINOv2 reached AUROC 0.8582, and model-derived suspicious-region localization reached macro IoU 0.4454 and Dice 0.5103. Supporting evidence was available for all 55 queries of an evaluated cohort, and audited pipeline latency was 23.40 ms (P95 28.30 ms) on a CPU testbed. We report limitations candidly, including specimen overlap across splits and synthetic ground truth.**

***Index Terms*—scanning electron microscopy, image retrieval, DINOv2, supervised contrastive learning, acquisition robustness, quality assessment, selective prediction, data curation.**

I. INTRODUCTION

Scanning electron microscopy (SEM) forms an image by rastering a focused electron beam over a specimen and recording secondary-electron (SE) or backscattered-electron (BSE) signals. The accelerating voltage sets the beam's interaction volume, and the detector sets which signal builds contrast, so the same specimen imaged at 5 kV with an SE detector and at 20 kV with a BSE detector can look strikingly different. Working distance, magnification, focus, stigmation, and charging add further variation. Scientific archives therefore hold collections in which how an image was acquired competes with what was imaged.

This creates four problems for repository-scale curation: (1) Acquisition-induced visual variance: embedding-space neighbors tend to share instrument settings rather than specimen identity; (2) Unverified metadata fusion: acquisition metadata is routinely assumed to help search, yet this is rarely tested against a strong visual backbone; (3) Redundancy: re-scans and re-exports inflate archives, but aggressive deduplication risks discarding legitimate acquisitions; (4) Absent auditability: automated suggestions need traceable provenance and a human decision point. We frame the task as scientific-image retrieval and quality-risk curation; it is not clinical diagnosis, and all quality indicators are image-derived and advisory.

Contributions: (i) An acquisition-aware supervised-contrastive adapter on frozen DINOv2 features reducing observed gap by 66.23% on held-out instrument. (ii) A quantified specialization trade-off between retrieval and artifact sensitivity, motivating dual representations without learned fusion. (iii) An image-derived quality-risk, suspicious-region localization, evidence-retrieval, and abstention path evaluated on controlled synthetic benchmarks. (iv) A reproducible curation platform with a 17-stage workflow, cryptographic provenance, and a documented negative result for late metadata fusion.

II. LITERATURE SURVEY

A. Micrograph retrieval: Early materials-informatics systems used handcrafted descriptors such as GLCM textures [1], SIFT [2], and bag-of-visual-features [3]; surveys of computer vision for microstructures are given in [4], and CBIR in [5]. In-domain pre-training such as MicroNet improves out-of-distribution segmentation [6].
B. Self-supervised foundations: Vision Transformers [7] trained with self-distillation [8], masked modeling [9], or DINOv2 [10] provide general features; ResNet [11] remains a common baseline.
C. Acquisition invariance: Contrastive objectives [12] and supervised contrastive learning [13] shape metric spaces; domain-divergence theory [14] motivates treating instrument shift as distribution shift.
D. Vector indexing: Exact inner-product search and graph approximate indices [15, 16] make 384-d embeddings searchable at interactive speed.
E. Quality, anomalies & abstention: No-reference quality [17], structural similarity [18], and anomaly benchmarks [19] inform our indicators; we use confidence abstention [20, 21] with calibration error [22].
F. Curation & FAIR: Perceptual hashing [23] and FAIR principles [24] underpin deduplication and provenance design.

III. THE SCI-INTEL PLATFORM

Fig. 1 depicts the workflow: login, upload, validation, storage, metadata, representation, quality screening, localization, retrieval, evidence retrieval, explanation, suggested action, uncertainty handling, review, curation, audit, and provenance. The two 384-d representations remain separate and are joined only by deterministic orchestration. The backend is FastAPI with JWT auth, SQLAlchemy, and a React/TypeScript workbench.



Fig. 1. SCI-INTEL architecture. Frozen DINOv2 executes quality screening and localization; the adapted representation executes acquisition-aware retrieval. Deterministic orchestration provides evidence without learned fusion.

Metric	Handcrafted	Frozen DINOv2	Adapted (3-seed)
Retrieval R@5 (Protocol U)	–	0.9858	0.9921
Retrieval MRR (Protocol U)	–	0.5200	0.5261
Acquisition gap Delta (lower better)	–	0.2016	0.0681
Risk AUROC (binary)	0.8281	0.8582	0.8230
Risk AUPRC (binary)	0.9808	0.9841	0.9792
11-class macro-F1	–	0.6837	0.6323
11-class balanced accuracy	–	0.7036	0.6645

TABLE I: Representation Specialization Trade-off Matrix.

IV. EXPERIMENTAL SETUP

Evaluation spans 6,085 active micrographs (Table II): HCCI SEM (774 PNGs), Carinthia SEM (4,591 JPEGs), and BBBC021 optical (720 TIFFs). The HCCI split is instrument-disjoint (train: Helios, 427; val: VEGA3, 135; test: Zeiss Gemini, 212) with zero image overlap. Because specimen classes overlap across partitions, the task is acquisition-robust retrieval, not unseen-specimen generalization. Protocol U evaluates 212 queries against 211 distractors with same-acquisition peers retained in the gallery. Synthetic benchmark evaluates 2,750 images across 11 classes, and localization evaluates 500 images with synthetic masks.

Dataset	Images	Format	Role
HCCI (SEM, cast iron)	774	PNG	Retrieval & synthetic benchmark
Carinthia (SEM, wafer)	4,591	JPEG	Cross-domain novelty screen
BBBC021 (bioimage)	720	TIFF (16-bit)	Ingestion & metadata profiling
Total Active Micrographs	6,085	Multi-format	Authoritative benchmark cohort
HCCI Train / Val / Test	427 / 135 / 212	PNG	Instrument-disjoint split (3 alloys shared)

TABLE II: Active Repository Datasets & Instrument Partitions.

V. RESULTS AND DISCUSSION

A. Acquisition Gap (Claim 1): The adapter shrinks Delta from 0.2016 to 0.0681 (66.23% reduction, 95% CI [65.72%, 67.19%], p = 5.03x10^-36, dz = 2.19, N=210 paired queries; Table III). Per-seed reductions are 60.76% (seed 42), 70.35% (seed 123), and 67.57% (seed 2024).

Representation	Within	Cross	Gap Delta [95% CI]	Red. (%)	dz
Frozen DINOv2	0.7811	0.5794	0.2016 [0.1833, 0.2096]	–	–
Adapter, seed 42	0.9055	0.8263	0.0791 [0.0713, 0.0823]	60.76	2.07
Adapter, seed 123	0.9129	0.8531	0.0598 [0.0530, 0.0631]	70.35	2.19
Adapter, seed 2024	0.9071	0.8417	0.0654 [0.0574, 0.0686]	67.57	2.26
Adapter, 3-seed mean	0.9085	0.8404	0.0681 [0.0606, 0.0713]	66.23	2.19

TABLE III: Acquisition-Geometry Gap Analysis (210 Paired Queries).

B. Retrieval (Claim 2): Table IV reports Protocol U. Adapter 3-seed mean attains R@1 0.1447, R@5 0.9921, and MRR 0.5261, versus 0.1321, 0.9858, and 0.5200 for frozen DINOv2. pHash attains P@5 0.6736 because relevance definitions utilize pHash thresholds.

Method	R@1	R@5	R@10	MRR	P@5
pHash-64	0.0991	0.9717	1.0000	0.4894	0.6736
dHash-64	0.1085	0.9481	0.9717	0.4995	0.6236
ResNet-50	0.0802	0.9811	1.0000	0.4836	0.5726
DINOv2 ViT-S/14	0.1321	0.9858	1.0000	0.5200	0.6160
Adapter, seed 42	0.1321	0.9953	1.0000	0.5230	0.6321
Adapter, seed 123	0.1462	0.9906	1.0000	0.5214	0.6217
Adapter, seed 2024	0.1557	0.9906	1.0000	0.5338	0.6443
Adapter, 3-seed mean	0.1447	0.9921	1.0000	0.5261	0.6327

TABLE IV: Protocol U Retrieval Benchmarks (212 Queries, 211 Candidates Each).

C. Late Metadata Fusion (Claim 3): Late score-level fusion yields delta R@1 = 0.0000 (alpha* = 1.0). Saturated visual features render late fusion redundant.

D. Duplicate Screening (Claim 4): SHA-256 revealed 0 exact duplicates; pHash flagged 483 groups, partitioned into 769 clusters (764 singletons, 5 review pairs).

E. Quality & Localization (Claim 5): DINOv2 attains AUROC 0.8582 and macro-F1 0.6837. Table V shows localization across 500 masks: macro IoU 0.4454 and Dice 0.5103.

Category	n	IoU	Dice	Precision	Recall
Charging-like (synthetic)	100	0.7794	0.8661	0.8812	0.8515
Overexposure	100	0.5621	0.6384	0.6841	0.5982
Underexposure	100	0.5579	0.6358	0.6795	0.5971
Local illumination	100	0.2515	0.3235	0.4210	0.2625
Clipping	100	0.0762	0.0880	0.2967	0.0822
Macro Average	500	0.4454	0.5103	0.5925	0.4983

TABLE V: Model-Derived Suspicious-Region Localization Precision (500 Synthetic Masks).

F. Evidence & Selective Prediction (Claim 6): Table VI reports selective prediction and cohort availability. Valid evidence was available for 100.0% of N=55 queries (2.0 items/query). Selective prediction reaches 99.08% accuracy at tau=0.4 (29.73% coverage) and 100% at tau=0.6 (9.73% coverage; ECE 0.3333).

Threshold tau	Coverage	Selective Acc.	Abstention	Cohort Parameter	Value
0.2	77.00%	78.28%	23.00%	Valid Evidence Availability	100.0%
0.4	29.73%	99.08%	70.27%	Same-Specimen Cross-Acq.	100.0%
0.6	9.73%	100.00%	90.27%	Cross-Instrument Evidence	100.0%
–	–	–	–	Zero Duplicate Evidence	0.0%
–	–	–	–	Mean Evidence Items / Query	2.0 (N=55)

TABLE VI: Selective Prediction Coverage-Accuracy & Evidence Availability Cohort (N=55).

G. Latency (Claim 7): Table VII reports audited serial CPU latency: 23.40 ms mean (P95 28.30 ms), tens of milliseconds per image.

Pipeline Execution Stage	Mean Latency (ms)	P95 Latency (ms)
Image Preprocessing & Hash Validation	0.35	0.48
Dual Representation Extraction (384-d)	3.12	3.95
Quality-Risk Screening	2.45	3.10
Spatial Suspicious-Region Localization	8.84	11.20
Comparable Evidence Retrieval (FAISS)	4.22	5.35
Evidence Aggregation & Explanation	4.42	5.60
Total Pipeline Execution (Serial CPU)	23.40 ms	28.30 ms

TABLE VII: Audited Serial Pipeline Latency Breakdown.

VI. LIMITATIONS AND THREATS TO VALIDITY

1. Specimen overlap: All splits share 3 alloy specimens; unseen-specimen generalization is not evaluated. 2. Synthetic quality labels: Degradations and masks are synthetically simulated. 3. Localization: Patch-residual saliency at 16-px resolution is heuristic. 4. Saturated metadata: Tested only at

score-level; deep cross-attention remains future work. 5. Evidence cohort: N=55 queries; expert diagnostic utility was not evaluated. 6. Latency: CPU execution dependent on benchmark hardware; GPU was unprofiled. 7. Specialization: Adapted space weakens artifact detection, necessitating dual representations.

VII. CONCLUSION AND FUTURE WORK

SCI-INTEL demonstrates that supervised contrastive adaptation reduces the observed acquisition gap by 66.23% on an unseen instrument, while frozen DINOv2 remains the superior artifact detector. Retaining dual representations under deterministic orchestration resolves the trade-off. Future work includes held-out alloys, cross-attention fusion, 3D FIB-SEM, and microscopist user studies.

COMPLIANCE WITH ETHICAL STANDARDS

This study utilized publicly available scientific imaging datasets (HCCI bearing steel, Carinthia SEM, and BBBC021 bioimage benchmark) and controlled synthetic augmentations. The research did not involve human subjects, patient data, clinical interventions, or live animal experiments, and therefore did not require institutional review board (IRB) approval.

DISCLOSURES & CONFLICTS OF INTEREST

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